

The number in the parentheses indicates the marks for that question

Instructions:

1. The exam-sheet consists of **9** questions. You are required to attempt **any 5** questions.
 2. Each question carries the marks indicated in parentheses.
 3. Use a separate answer-sheet for answering the questions.
 4. You may assume that the conditions for all problems are ideal. If you notice there is inadequate information provided for a question, you can state your assumptions and answer the question.
 5. This is a closed-book test. You are not permitted to refer to any notes, books or online resources.
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1. ✓ (a) With a neat and labeled diagram, explain the **OSI model** and **data encapsulation** process in computer networks. (8)
 - (b) Draw the **IPv4 header** and explain all its fields. Include details such as version, header length, type of service, total length, identification, flags, fragment offset, time to live (TTL), protocol, header checksum, source IP address, destination IP address, and options. (9)
 - (c) What is the difference between **telecommunication** and **networking**? Explain their main characteristics and purposes. (3)
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2. (a) Briefly explain IPv6 addressing and expressions. (10)
 - (b) What is meant by shortened expression in IPv6? (3)
 - (c) Explain **Global Unicast Addresses** in IPv6. (3)
 - (d) Discuss the significance of the following reserved address blocks in IPv6:

- 2000::/3
- FC00::/7
- FE80::/10
- FF00::/8

(4)

3. (a) Name the appropriate OSI model layer or device (such as a hub, switch, access point, router, etc.) for each of the following description:.

1. This device sends and receives information about the Network layer.
2. This layer creates a virtual circuit before transmitting between two end stations.
3. A layer-3 switch or multilayer switch.
4. This device uses hardware addresses to filter a network.
5. Ethernet is defined at these layers.
6. This layer supports flow control and sequencing.
7. This device can measure the distance to a remote network.
8. Logical addressing is used at this layer.
9. Hardware addresses are defined at this layer.
10. This device creates one big collision domain and one large broadcast domain.
11. This device creates many smaller collision domains, but the network is still one large broadcast domain.
12. This device can never run full duplex.
13. This device breaks up collision domains and broadcast domains.

(13)

(b) Draw the **TCP header** and explain all its fields. Include details such as source port, destination port, sequence number, acknowledgment number, data offset, reserved bits, flags, window size, checksum, urgent pointer, and options.

(7)

4. (a) Find the **subnet**, **broadcast address**, and **valid host range** for Question A through Question F, and answer Questions G–J in a few sentences. Show all the steps of your calculation for all the parts of the question.

- A. 192.168.100.25/30
- B. 192.168.100.37/28
- C. 192.168.100.66/27
- D. 192.168.100.17/29
- E. 192.168.100.99/26
- F. 192.168.100.99/25
- G. You have a Class B network and need 29 subnets. What is your subnet mask?
- H. What is the broadcast address of 192.168.192.10/29?
- I. How many hosts are available with a Class C /29 mask?
- J. What is the subnet for host ID 172.16.3.65/23?

(20)

5. (a) Why do we need **IPv6**? Explain its significance and advantages over IPv4.

(8)

- (b) A sender needs to transmit 10 bytes to a receiver using the **Go-Back-N** protocol. The window size is 3, and every 5th packet is lost during transmission. No ACK from the receiver are lost.

- Calculate the total number of bytes that will be transmitted, including retransmissions, until all 10 bytes are successfully received.
- Explain your reasoning and show all steps of the calculation.

(12)

6. (a) What are some **future applications of networking**? Discuss emerging trends and technologies that may shape the future of computer networks.

(6)

- (b) What is the need for **Network Function Virtualization (NFV)** and **Software-Defined Networking (SDN)** in modern computer networks?

In your answer, you can explain the challenges in traditional networks that NFV and SDN aim to address. Discuss the benefits and advantages of implementing NFV and SDN, such as scalability, flexibility, and cost-efficiency. Provide examples of use cases where NFV and SDN improve network management and performance.

(8)

- (c) List some differences between **TCP** and **UDP**. Discuss their features, reliability, connection type, and typical applications.

(6)

7. (a) What is the need for an **Access Point, Router, and Switch** in a computer network? With the help of a neat and labeled diagram:

- Explain the role and function of an **Access Point** in providing wireless connectivity.
- Describe how a **Router** connects different networks and manages data traffic.
- Discuss the purpose of a **Switch** in connecting devices within a LAN and forwarding data frames efficiently.
- Include examples of scenarios where each device is essential.

(10)

(b) List the characteristics of **fiber optic communications** and explain their advantages in computer networks. Include aspects such as bandwidth, signal attenuation, interference, and security.

(5)

(c) How long does it take a packet of length 1,000 bytes to propagate over a link of distance 2,500 km, with a propagation speed of (2.5×10^8) meters/second, and a transmission rate of 2 Mbps? More generally, how long does it take a packet of length L bits to propagate over a link of distance d meters, propagation speed s (in meters/second), and transmission rate R (in bits per second)? Does this delay depend on packet length L ? Does this delay depend on transmission rate R ?

(5)

8. For each of the following questions, answer in 2-3 sentences.

- (a) Which layer of the OSI model is responsible for end-to-end delivery of data? (2)
- (b) Mention any two functions of the Data Link Layer. (2)
- (c) What is the significance of subnet mask in IPv4 addressing? (2)
- (d) What are some frequency bands typically used in Wi-Fi communication? (2)
- (e) Name a few parameters used to measure network performance. (2)
- (f) What is the maximum transmission unit (MTU) in Ethernet? (2)
- (g) Define the term **protocol** in the context of computer networks. (2)
- (h) What is the main function of the Domain Name System (DNS)? (2)
- (i) Define the range of IPv6 addresses and provide an example of IPv6 address. (2)

(j) How many bits are used in an IPv4 address? Mention the range of IPv4 address. (2)

9. For each of the following questions, select the correct answer(s) from the given options. Some questions may have one or more correct answers. Mark all that apply. For each question, explain/justify in 2-3 sentences why the selected option is correct and other options are incorrect. For each question, 0.5 marks are for selecting the correct option and 1.5 marks are for providing appropriate explanation/justification.

(a) Which of the following is an **invalid IP address** for a host?

- A. 10.0.0.1
- B. 128.0.0.1
- C. 224.0.0.1
- D. 172.0.0.1

(0.5 + 1.5)

(b) Which of the following is a disadvantage of the **star topology**?

- A. When a port on the central concentrating device fails, the attached end device loses connectivity to the rest of the network.
- B. When the central concentrating device experiences a complete failure, all attached devices lose connectivity to the rest of the network.
- C. In a star topology, a more expensive type of host must be used when compared to the host used when implementing a physical bus.
- D. It is more difficult to add stations and troubleshoot than with other topologies.

(0.5 + 1.5)

(c) Host 1 sent a **SYN** packet to Host 2. What will Host 2 send in response?

- A. ACK
- B. SYN-ACK
- C. SYN-NAK
- D. SYN

(0.5 + 1.5)

- (d) You are connected to a server on the Internet and you click on a link on the server and receive a time-out message. What layer could be the cause of this message?
- A. Application
 - B. Transport
 - C. Network
 - D. Physical

(0.5 + 1.5)

- (e) After rebooting an Access Point (AP), a user is no longer able to connect to the enterprise LAN. A technician plugs a laptop into the same network jack and receives the IP address 169.254.0.200. Which of the following is the **MOST** likely cause of the issue?
- A. DHCP scope exhaustion
 - B. Signal attenuation
 - C. Channel overlap
 - D. Improper DNS configuration

(0.5 + 1.5)

- (f) What protocol helps devices share the bandwidth evenly without having two devices transmit at the same time on the network medium?
- A. TCP/IP
 - B. CSMA/CD
 - C. HTTPS
 - D. TFTP

(0.5 + 1.5)

- (g) A technician has deployed multiple temperature monitoring devices with IP addresses in the 10.137.24.0/25 network. Such devices must receive an ICMP keep-alive packet from the server every five minutes. Which of the following destination IP addresses must the technician configure on the server to send keep-alive packets to all the devices?
- A. 10.137.24.0
 - B. 10.137.24.1
 - C. 10.137.24.127
 - D. 10.137.24.128

(0.5 + 1.5)

(h) Which of the following allows a server to distinguish among different simultaneous requests from the same host?

- A. They have different port numbers.
- B. A NAT server changes the IP address for subsequent requests.
- C. A server is unable to accept multiple simultaneous sessions from the same host. One session must end before another can begin.
- D. The MAC address for each one is unique.

(0.5 + 1.5)

(i) You have a network with a subnet of 172.16.17.0/22. Which are the valid host addresses?

- A. 172.16.17.1 255.255.255.252
- B. 172.16.0.1 255.255.240.0
- C. 172.16.20.1 255.255.254.0
- D. 172.16.16.1 255.255.255.240
- E. 172.16.18.255 255.255.252.0
- F. 172.16.0.1 255.255.255.0

(0.5 + 1.5)

(j) Your router has the following IP address on Ethernet0: 172.16.2.1/23. Which of the following can be valid host IDs on the LAN interface attached to the router? (Select two.)

- A. 172.16.0.5
- B. 172.16.1.100
- C. 172.16.1.198
- D. 172.16.2.255
- E. 172.16.3.0
- F. 172.16.3.255

(0.5 + 1.5)